

Unfortunately, neither of these models describes the world perfectly, and sometimes actual skews observed in the market are more complicated than either of these two models suggests. To complicate things further, the actual skew patterns observed in the market have tended to change over the course of time. The market has come up with colorful names to describe various skew patterns that have been observed from time to time. For example, two skew patterns that have existed at various times in the interest rate options market are often referred to as *supernormal* and *smirk*. These two skew patterns are depicted in Figure 5.9 with the skew pattern implied by the normal model superimposed. When the volatility skew is supernormal, lognormal implied volatilities lie everywhere above the levels implied by the normal model. And when a smirk is observed in the market, volatilities for very high-strike options are actually above vols of at-the-moneys.

The Importance of Understanding Skew

The understanding of skew dynamics is more than just of theoretical interest, as the following example will illustrate.

Example

Consider again our stylized example, the swap curve in Table 5.2 and the lognormal implied matrix in Table 5.1. Suppose that a dealer is asked to bid (i.e., to buy) \$1bn of 2 into 3 payer swaptions struck *50 out* (i.e., the strike is 50 bps above the at-the-money forward strike in the case of a payer swaption) from a client. Assume that the trader uses Black's model.

The first step to pricing up this swaption is to compute the 2x5 swap rate. As shown in the top half of Figure 5.10, we find that the 2x5 swap rate is 5.976%. Thus, the strike on the payer swaption is 6.476%. According to the vol grid in Table 5.1, 2 into 3 vol for swaptions struck at-the-money forward is 26.5%. Suppose that the trader either forgets or is unaware of the existence of skew and uses this vol to put a bid on the swaption. As shown in the bottom half of Figure 5.10, the price at which the dealer will bid the option is 168.8 bps, or about \$16,877,000 on a notional of \$1bn. Assume that soon after buying the swaption from the client, rates have not changed, and the dealer goes out into the market to try to sell the option in order to close out his position. Upon looking for bids in the market, he finds that other market participants are pricing the option with skew. In particular, the best bid that he can find in the market is 26.25% lognormal implied vol. In a calculation not shown here, this equates to a bid of 166.7 bps, or about \$16,672,000 on a notional of \$1bn. Thus, by not incorporating skew into his pricing, the trader has lost about \$205,000. □

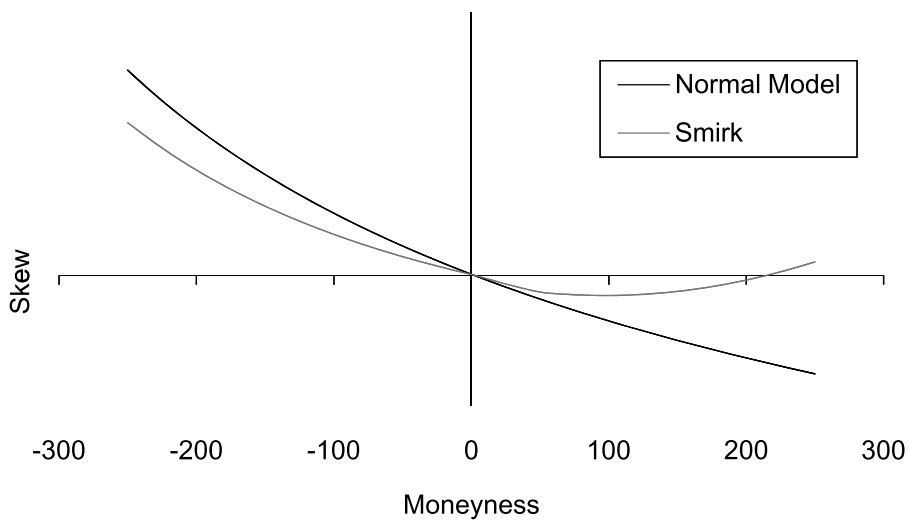
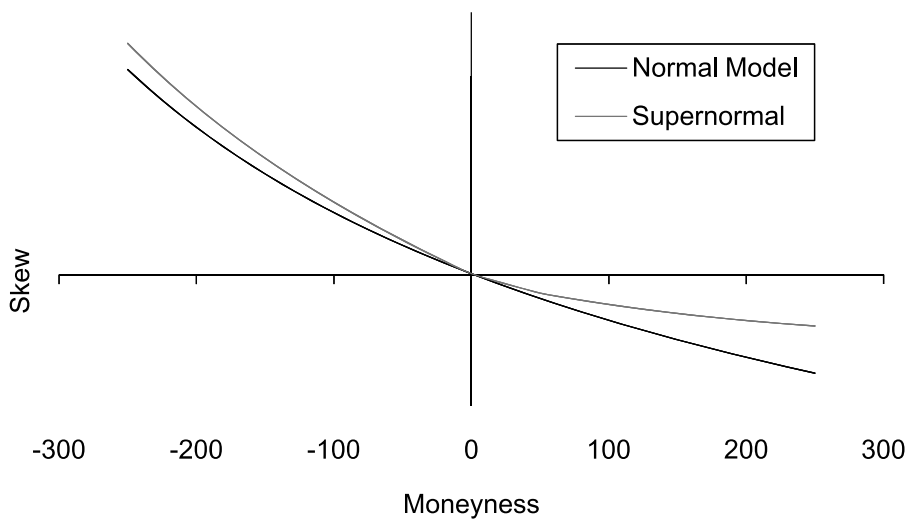


Figure 5.9: Some Volatility Skews Observed in the Swaptions Market: Empirical Lognormal Implied Vols Sometimes Observed in the Market with Theoretical Lognormal Implied Vols from the Normal Model Superimposed

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	0	0	0	0
2	5.113%	5.226%	0	0	0	0
3	5.312%	5.712%	5,975,505	5,116,090	5,712,202	4,890,656
4	5.494%	6.041%	5,975,505	4,824,617	6,041,364	4,877,792
5	5.635%	6.202%	5,975,505	4,542,865	6,202,087	4,715,123
				14,483,572		14,483,572
Fixed rate			5.976%	NPV	0	
Notional			100,000,000			

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.0270
2	5.110%	5.113%	1x2		d_2	-0.4018
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074	$FN(d_1) - XN(d_2)$	0.006963
5	5.600%	5.635%	4x5	0.7602	Payer swaption (in bps)	16,876,810 168.8
F	5.976%		$A =$	2.4238		
σ	26.5%					
T	2		Notional	1,000,000,000		
X	6.476%					

Figure 5.10: Pricing Up the 2 into 3 Payer Swaption Struck 50 Out but Using Vol for the At-the-Money Forward Strike

Why Skew?

The Black-Scholes framework suggests that if we allow the strike of an option to vary, the lognormal implied vol of the option should be the same. But as we just discussed, this is far from the case in the interest rate option markets in which we often observe a downward-sloping skew. Why should such skew exist?

It is somewhat incontrovertible to say that any skew pattern that is observed in the market is the result of the supply and demand effects for options of particular strikes. The traditional structural line-up of buyers and sellers in the vol markets is complex. Figure 5.11 gives a sense of the various players that have traditionally either bought or sold options in the interest rate options market. In this figure, an arrow at the end of a line pointing to an entity indicates that the entity represented has traditionally been a net buyer of options. Thus, for example, mortgage servicers have traditionally been net buyers of low-strike receivers and CMS floors. Corporations and

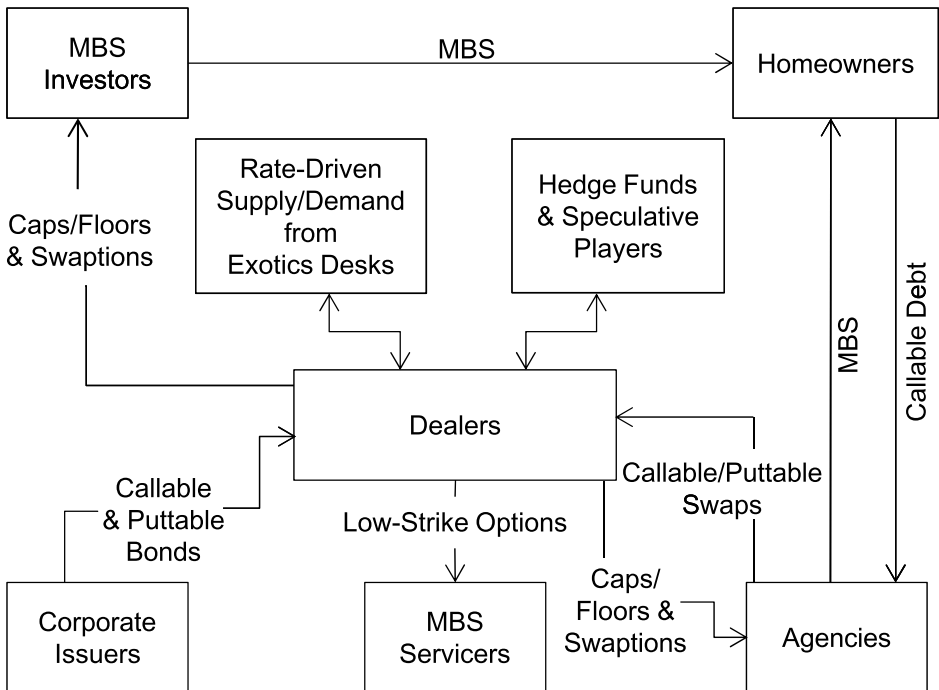


Figure 5.11: Supply and Demand in the Interest Rate Option Markets

the U.S. agencies have been traditional suppliers to the market of high-strike options in the form of various callable and puttable structures.²⁶ Hedge funds have traditionally been both buyers and sellers of options, as indicated by the arrows in both directions.

As this figure indicates, the supply and demand factors governing the vol markets is quite complicated, with a variety of players involved. Despite this complexity, the conventional wisdom in the market is that there tends to be net demand from customers for low-strike options. This demand is reflected in the higher implied vols at which these options trade. The demand for these low-strike options comes from a variety of buy-side entities, many of whom are hedging against the risk of mortgage prepayment.²⁷

It is also conventional wisdom that there has traditionally been a supply of high-strike options into the market. Entities such as corporations and the agencies have traditionally issued fixed-rate callable bonds. As we will discuss in Chapter 6, these issuers often swap these bonds, thereby transforming the nature of their option-embedded liabilities from fixed to floating and, in the process, wind up selling high-strike options to dealers. This is not to say that there is no buying of high-strike options from other buy-side entities. There are certainly a number of institutions that have tended to buy high-strike options.²⁸ But this buying has often been tempered by a decent supply of high-strike options.

Expectations of what may happen to implied vol if certain rate levels are obtained may also be a determinant of skew. For instance, consider an option dealer who sells some low-strike options to one of its clients. In order to hedge the position, the dealer may wish to buy some options. Off-the-money options are not as liquid as at-the-money options, and as such the dealer may be forced to buy some at-the money options to hedge.²⁹ Suppose that the market does rally after the fact. In this case the low-strike options that the dealer sold are nearing the money now (and therefore now have significant vega), and the at-the-money options that were bought as a hedge are now out-of-the-money (and therefore now have de minimus vega). In this case, the dealer will find himself short vol. In order to rebalance, the dealer will have to buy some options. Thus, if the market rallies and the strike on

²⁶We will look at some of these structures in Chapter 6.

²⁷See Appendix D for a brief review of mortgages and the risk of prepayment of mortgage securities.

²⁸In fact, the buying of high-strike options is sometimes so pronounced in the market that such options trade at an implied volatility higher than that of at-the-money strikes. This corresponds to the smirk pattern of implied volatilities shown in Figure 5.9, which is sometimes observed in the market.

²⁹Such a hedge is known to create *strike mismatch*, in which the options one is long and that one is short have different strikes. Managing strike mismatch can often be a challenging feat for option traders.

the dealer's short option position becomes at-the-money, the dealer knows that he will have to buy vol, which will on the margin tend to raise the implied vols at which he can buy options (i.e., the marginal option buying from the dealer will put upward pressure on option prices in the market). This may persuade him to initially sell the low-strike options at higher levels of implied vol.

And of course, another explanation for the existence of skew is that the distribution of interest rates assumed under Black's model may just be wrong. Black's model assumes that interest rates are lognormally distributed. As we discussed in Section 5.3.1, there is good reason to doubt the appropriateness of the assumption of lognormality for interest rates. As we discussed, there is empirical evidence to suggest that a better distributional assumption for interest rates is normality. And, as we discussed, if the normal model is assumed to be correct (i.e., if interest rates are normally distributed), then (as we will demonstrate in Section 5.4.5) the pattern of lognormal implied vols must exhibit downward skew.

5.4 The Normal Model

With a significant amount of doubt cast over the appropriateness of Black's model for the pricing of interest rate options, the question is, What option pricing model is more appropriate? In this section, we explore in detail the concept of the *normal model*, in which rates are assumed to be normally distributed, as an alternative.

5.4.1 Background

Of the three shortcomings of Black's model for the pricing of interest rate options discussed in Section 5.3, the existence of skew is perhaps the most troubling and the one that provided ample motivation to seek an alternative model that explained skew. But prior to the mid-1990s, the concept of skew was not well recognized in the industry. Many swaption traders did not focus on skew to any great extent when pricing options. Many were undoubtedly unaware of the existence of skew. Others may have just had their hands full trying to price up at-the-moneys without the distraction of entertaining the notion of skew. And still there were others who used various "fudge factors" when pricing up off-the-money options in an attempt to adjust for skew.

At the same time, transparency in the vol markets was limited and bid/offers were wide, contributing to further difficulties in the monitoring and analysis of skew. But over time, people who worked in the vol markets became more and more aware of the existence of skew, and its study and understanding became of paramount importance. Given the existence of skew,

and the fact that Black's model did not explain skew at all (Black's model assumes, of course, flat skew—that lognormal implied vols should not vary with strike), the challenge became finding an option pricing model that did explain lognormal implied skew that was empirically observed and ideally one whose own implied volatility parameter did not vary with strike.

5.4.2 The Model

As we discussed briefly in Section 5.3.1, in the normal model we assume that interest rates are normally distributed. In particular, we assume that a swap rate R to be observed at time T is normally distributed. That is, we assume

$$R \sim N(\mu, \sigma_N^2 T), \quad (5.20)$$

where μ and $\sigma_N^2 T$ are the mean and variance of the distribution, respectively. The parameter σ_N is referred to as *normal vol* or *normalized vol* or *basis point vol* or *annualized basis point vol*. The distributional assumption made in Equation 5.20 is referred to as the normal model. Under the normal model, σ_N is assumed to be a constant that does not vary with the level of rates nor with the strike of the option. In the normal model, σ_N is a measure of *absolute volatility*. Whereas the parameter σ in Black's model is expressed as a percentage, σ_N in the normal model is expressed as an absolute number of basis points.

The distributional assumption made on swap rates under the normal model is depicted in Figure 5.12. There are two things worth noting about the distributional assumption made in the normal model. First, the model makes a statement only of the terminal distribution of the swap rate. That is, the assumption being made is that the swap rate R at time T is normally distributed. The model says nothing about the distribution of the underlying prior to time T . Second, the normal model does allow for the possibility of negative interest rates.³⁰

Example

Suppose that the 10-year swap rate in one year is normally distributed with $\mu = 5\%$ and $\sigma_N = 120$ bps. Then under the normal model, there is approximately a 68.2% chance that the 10-year swap rate in a year is between $5\% - 120 \text{ bps} = 3.80\%$ and $5\% + 120 \text{ bps} = 6.20\%$. There is approximately a

³⁰Some would criticize the normal model for admitting the possibility of negative rates (a criticism that may be less robust in recent times during which some interest rates have gone negative). Others would just “live with it,” believing that the probability of rates going negative under most meaningful parameterizations of the model is small enough to safely be ignored. Still others would look for more sophisticated models altogether.

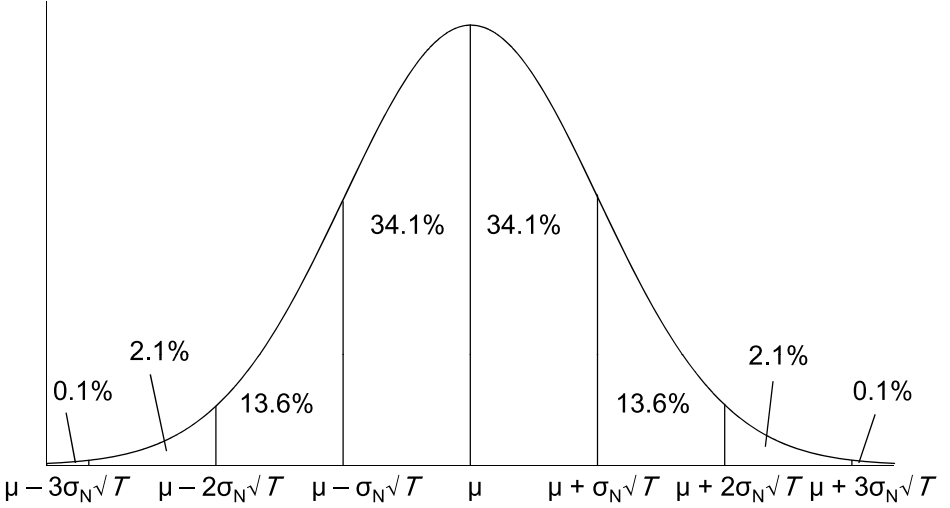


Figure 5.12: Distribution of Swap Rates Under the Normal Model

95.4% chance that the 10-year swap rate in a year is between $5\% - (2 \times 120 \text{ bps}) = 2.60\%$ and $5\% + (2 \times 120) \text{ bps} = 7.40\%$. Under the normal model, there is a 0.00155% chance that the 10-year swap rate in a year is negative.³¹ $\square$

5.4.3 Pricing Under the Normal Model

Consider again our stylized example with annual bond fixed coupons versus 1-year LIBOR. In Appendix E, we show that under the normal model the price of a T into n payer swaption struck at X is

$$A\sigma_N\sqrt{T} \left[\frac{1}{\sqrt{2\pi}} e^{-d^2/2} + dN(d) \right], \quad (5.21)$$

³¹In order to determine the probability that R is less than zero in a year, define a random variable x by

$$x = \frac{R - 5\%}{0.012}$$

and note that x is a standard normal (i.e., has mean zero and variance one). The probability that R is negative in a year is equal to the probability that

$$x < \frac{0\% - 5\%}{0.012}.$$

This probability is easily determined using spreadsheet software or through a standard normal table of probability distribution found in, for example, Ross (2009).

where

$$d = \frac{F - X}{\sigma_N \sqrt{T}} \quad (5.22)$$

and

$$A = \sum_{t=T+1}^{T+n} DF(0, t) \quad (5.23)$$

(the annuity factor) where

$$DF(0, t) = \frac{1}{[1 + zc(t)]^t}, \quad (5.24)$$

F is the forward swap rate, T is the time to expiration, σ_N is normal implied vol and $N(\cdot)$ denotes the cumulative normal distribution function.

Similarly, under the normal model, the value of a T into n receiver swaption struck at X is given by

$$A\sigma_N\sqrt{T} \left[\frac{1}{\sqrt{2\pi}} e^{-d^2/2} - dN(-d) \right]. \quad (5.25)$$

By setting the strike X equal to F and adding Equations 5.21 and 5.25, the value of a swaption straddle struck at-the-money forward is³²

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A. \quad (5.26)$$

To find the price of these options in dollars, we would multiple Equations 5.21, 5.25, and 5.26 by the notional of the trade.

The expression for a swaption straddle struck at-the-money forward in Equation 5.26 is very commonly used throughout the industry. In Equation 5.26, we see that under the normal model the value of an at-the-money swaption straddle is linear both in the square root of time and in normalized implied vol. Note that in the normal model the value of an at-the-money swaption straddle does not depend on the level of the underlying forward swap rate.³³

³²The quantity $\sqrt{2/\pi}$ is approximately 0.8, so sometimes the expression in Equation 5.26 is written as $\sqrt{T} \times 0.8 \times \sigma_N \times A$.

³³This, while at first seemingly surprising, should jibe with our intuition. Under the normal model, the level of volatility does not depend on the level of rates, and thus the normal implied volatility σ_N “tells us everything we need to know” about the possible dispersion of rates in the model. By comparison, under the lognormal model, the concept of volatility is a relative one, and thus it is not surprising that the level of the underlying forward rate, F , enters the equation of a swaption straddle struck at-the-money forward, as shown in Equation E.14 of Appendix E.

Example

Consider again our stylized example and the swap curve given in Table 5.2. Let's price a 1 into 3 payer swaption and a 1 into 3 receiver swaption each struck "100 out" (meaning 100 bps above and below, respectively, the at-the-money forward strike) using the normal model. Let us assume that normal implied vol, σ_N , for each of these options is 112.7 bps. Assume a notional of \$25mm.

In accordance with Equations 5.21 and 5.25, we will need as an input in the pricing the underlying forward swap rate. In Section 3.3.4 we showed that the 3-year swap rate, one year forward was equal to 5.644%. Thus, the strike on the 1 into 3 payer swaption struck 100 out is 6.644%, and the strike on the 1 into 3 receiver swaption struck 100 out is 4.644%. We have also previously found in Section 5.1.6 that the value of the annuity factor over the life of the underlying swap, A , is 2.5687. For the payer swaption, we have

$$d = \frac{F - X}{\sigma_N \sqrt{T}} = \frac{0.05644 - 0.06644}{0.01127 \times 1} = -0.8873.$$

From Equation 5.21, the price of the 1 into 3 payer swaption struck at 6.644% is

$$\begin{aligned} & A\sigma_N\sqrt{T} \left[\frac{1}{\sqrt{2\pi}} e^{-d^2/2} + dN(d) \right] \\ &= 2.5687 \times 0.01127 \times \sqrt{1} \\ &\quad \times \left[\frac{1}{\sqrt{2\pi}} e^{-0.8873^2/2} - 0.8873 \times N(-0.8873) \right] \\ &= 0.00298, \end{aligned}$$

or 29.8 bps of the notional. On a notional of \$25mm, the premium is \$25mm $\times$ 29.8 bps = \$74,500.

For the receiver swaption, we have

$$d = \frac{F - X}{\sigma_N \sqrt{T}} = \frac{0.05644 - 0.04644}{0.01127 \times 1} = +0.8873.$$

From Equation 5.25, the price of the 1 into 3 receiver swaption struck at 4.644% is

$$\begin{aligned} & A\sigma_N\sqrt{T} \left[\frac{1}{\sqrt{2\pi}} e^{-d^2/2} - dN(-d) \right] \\ &= 2.5687 \times 0.01127 \times \sqrt{1} \\ &\quad \times \left[\frac{1}{\sqrt{2\pi}} e^{-0.8873^2/2} - 0.8873 \times N(-0.8873) \right] \\ &= 0.00298, \end{aligned}$$